Generate terms from a position rule

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: Find the first five terms of 4n - 1. Copy or complete the first step.

2. Continue 7, 12, 17, 22 and state the rule.

3. Find the missing terms: 84, __, 62, __, 40.

4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the first five terms of 4n - 1. Copy or complete the first step.
   Answer 3, 7, 11, 15, 19.

2. Continue 7, 12, 17, 22 and state the rule.
   Answer 27, 32; add 5.

3. Find the missing terms: 84, __, 62, __, 40.
   Answer 73 and 51.

4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.
   Answer State what the variable represents, substitute its value and preserve the operation structure.

Generate terms from a position rule

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ___________________

1. Find the first five terms of $4n - 1$.

2. Continue 7, 12, 17, 22 and state the rule.

3. Find the missing terms: 84, __, 62, __, 40.

4. Solve this independently and show every stage: Find the first five terms of $4n - 1$.

5. Use a second method or representation for: Continue 7, 12, 17, 22 and state the rule.

6. Check this result using an inverse, estimate or known fact: Find the missing terms: 84, __, 62, __, 40.

7. Prove or explain your reasoning: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Find the missing terms: 84, __, 62, __, 40.

Teacher answers and checking prompts

1. Find the first five terms of 4n - 1.
Answer 3, 7, 11, 15, 19.

2. Continue 7, 12, 17, 22 and state the rule.
Answer 27, 32; add 5.

3. Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51.

4. Solve this independently and show every stage: Find the first five terms of 4n - 1.
Answer 3, 7, 11, 15, 19.

5. Use a second method or representation for: Continue 7, 12, 17, 22 and state the rule.
Answer Expected result: 27, 32; add 5. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Find the missing terms: 84, __, 62, __, 40.
Answer Expected result: 73 and 51. A valid check must be shown.

7. Prove or explain your reasoning: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer C = 6n + 4. The explanation must show why the method works.

8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.

9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Find the missing terms: 84, __, 62, __, 40.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Generate terms from a position rule

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

2. Prove the answer to this question, rather than only calculating it: Find the first five terms of 4n - 1.

3. Solve this in two different ways and compare them: Continue 7, 12, 17, 22 and state the rule.

4. Work backwards from the answer to reconstruct the question: 73 and 51.

5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.

6. Create a new example that tests the same idea as: Find the missing terms: 84, __, 62, __, 40.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
   Answer C = 6n + 4. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Find the first five terms of 4n - 1.
   Answer Expected result: 3, 7, 11, 15, 19. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Continue 7, 12, 17, 22 and state the rule.
   Answer Expected result: 27, 32; add 5. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 73 and 51.
   Answer One valid reconstruction is: Find the missing terms: 84, __, 62, __, 40. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.
   Answer State what the variable represents, substitute its value and preserve the operation structure.

6. Create a new example that tests the same idea as: Find the missing terms: 84, __, 62, __, 40.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.